

UNIVERSITY OF SCIENCE AND TECHNOLOGY AT ZEWAIL CITY



Machine Learning Corrections to Nuclear Density Functional Theory Predictions

Omar Ziada 202200872

Hazem Mohamed 202200777

Supervised by: Sir Hisham Anwer
Dr. Mohamed Tolbla
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Abstract

This project explores the application of Machine Learning (ML) techniques to improve theoretical predictions in nuclear physics. Using the DRHBc (Deformed Relativistic Hartree–Bogoliubov theory in Continuum) dataset for even- Z nuclei ($N = 5437$), we implement a physics-informed Δ -learning approach to predict experimental binding energies (E_b) and charge radii (R_{ch}). The study addresses the discrepancy between theoretical calculations and experimental observations by training regression models to learn the residual errors ($\Delta = \text{Exp} - \text{Theory}$). We perform extensive data pre-processing, including the parsing of quantum numbers and physics-based imputation of separation energies. Several algorithms, including Linear Regression, Ridge, Lasso, Random Forest, XGBoost, and Support Vector Regression (SVR), are evaluated. The results demonstrate that tree-based ensemble methods (XGBoost and Random Forest) significantly outperform linear models in capturing non-linear nuclear effects such as shell closures and deformation. Additionally, classification models were employed to accurately distinguish between spherical and deformed nuclei. The final output is a comprehensive, imputed dataset that enforces nuclear additivity constraints. Updated analyses include refined statistical comparisons and enhanced pattern discovery in the R_{ch} vs. E_b correlations.

Introduction

1.1 Background and Objectives

The accurate description of nuclear properties, such as binding energy and charge radius, is fundamental to understanding the synthesis of elements in the universe and the structure of matter. Theoretical models like Density Functional Theory (DFT) often deviate from experimental values due to approximations in the energy density functional.

This report details a Machine-learning framework to correct these systematic errors. The primary objectives are to perform rigorous data analysis, implement physics-informed feature engineering, and develop ML models that predict the residual difference between theory and experiment.

Dataset Description

The dataset consists of 5437 entries representing even-Z nuclei. It includes both experimental measurements (denoted with ‘_exp’) and theoretical predictions from the DRHBc theory (denoted with ‘_cal’). The specific variables included in the analysis are detailed in Table 1.

Table 1: Description of Dataset Variables

Column	Units	Description
Z	Integer	Proton number
N	Integer	Neutron number
A	Integer	Mass number
Eb_cal	MeV	Binding energy from DRHBc calculations
Ebrot_cal	MeV	Binding energy plus rotational correction
Eb_exp	MeV	Experimental binding energy
S2n	MeV	Two-neutron separation energy
S2p	MeV	Two-proton separation energy
Sn	MeV	One-neutron separation energy
Rn	fm	Neutron rms radius
Rp	fm	Proton rms radius
Rm	fm	Matter rms radius
Rch_cal	fm	Charge radius from DRHBc calculations
Rch_exp	fm	Experimental charge radius
beta_n	-	Neutron quadrupole deformation
beta_p	-	Proton quadrupole deformation
beta_t	-	Matter quadrupole deformation
lam_n	MeV	Neutron Fermi surface
lam_p	MeV	Proton Fermi surface
$m^{\pi}(N)$	-	Quantum numbers m^{π} of the blocked neutron orbital

EDA (Exploratory Data Analysis)

Initial inspection revealed missing values (denoted as ‘-’) in experimental columns, necessitating robust imputation strategies. Statistical descriptions provided insight into the range of binding energies (approx. 1000–1100 MeV in the sample subset) and radii (approx. 5–6 fm).

3.1 Distributions

Histograms with Kernel Density Estimation (KDE) were generated to visualize the distribution of observables. The distribution of deformation parameters (β) highlights the existence of both spherical (near $\beta = 0$) and deformed nuclei.

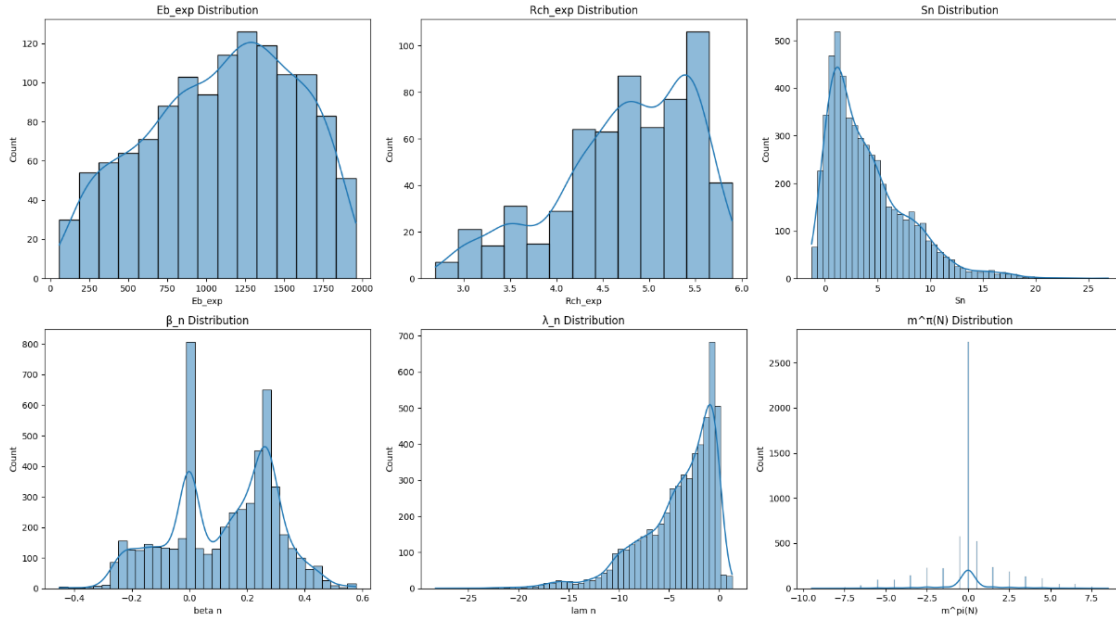


Figure 1: Histograms showing the distribution of binding energies, radii, separation energies, and deformation parameters.

3.2 Correlations

A correlation matrix was computed to identify redundancies. As expected, theoretical binding energy (E_{cal}) is highly correlated with rotational energy (E_{rot}), suggesting redundancy. Strong correlations were also observed between A and R_{ch} , following the $R \propto A^{1/3}$ law.

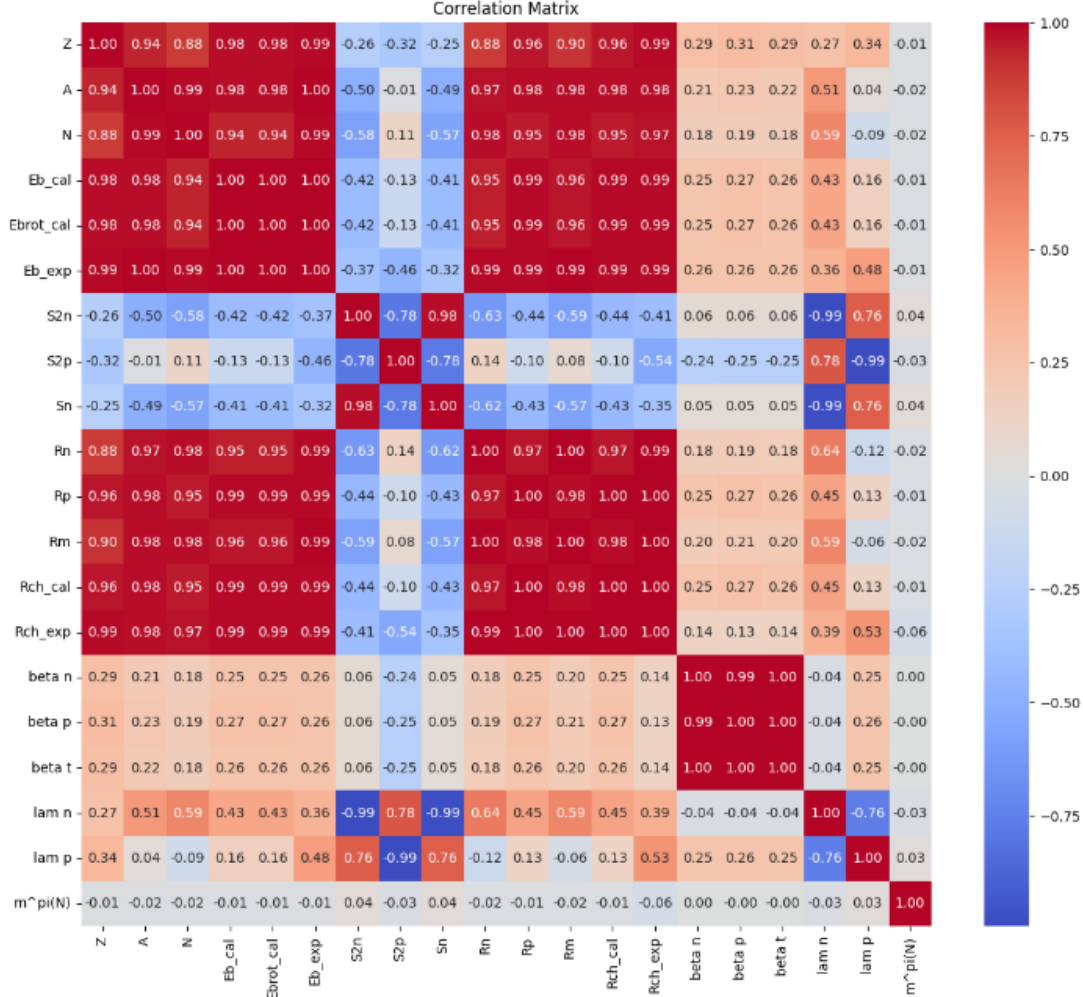


Figure 2: Heatmap displaying the correlation coefficients between nuclear features.

Data Preprocessing Steps

4.1 Data Cleaning

Non-numeric entries (hyphens) were converted to ‘NaN’. A crucial preprocessing step involved parsing the quantum number column $m^\pi(N)$. This column contained string representations of angular momentum and parity (e.g., "5/2+").

The parsing logic implemented was:

1. Extract the parity (+ or −) and convert to sign (+1, −1).
2. Parse the fraction (e.g., "5/2") into a float (2.5).
3. Combine to form a continuous numeric feature: $m_{num}^\pi = j \times \text{parity}$.

4.2 Physics-Informed Feature Engineering

To enable the ML models to learn nuclear structure effects, several domain-specific features were engineered:

1. Isospin Asymmetry (δ): Derived from the liquid drop model, this term captures the energy penalty for unequal numbers of protons and neutrons:

$$\delta = \frac{N - Z}{A} \quad (1)$$

2. Geometric Scaling ($A^{1/3}$): Since nuclear volume scales with A , the radius scales with $A^{1/3}$. This is a linear predictor for R_{ch} .

3. Proximity to Magic Numbers: Shell effects cause sudden jumps in binding energy at specific "magic" numbers ($M = \{2, 8, 20, 28, 50, 82, 126\}$). We calculated the distance to the nearest magic number for both protons and neutrons:

$$D_Z = \min_{m \in M} |Z - m|, \quad D_N = \min_{m \in M} |N - m| \quad (2)$$

4. Neutron Parity (P_N): A binary feature capturing even-odd effects in pairing energy. The pairing term significantly influences binding stability, favoring even-even nuclei over odd-odd ones.

$$P_N = N \pmod{2} \quad (3)$$

Model Design & Justification

This chapter details the two distinct machine learning frameworks developed in this study: the direct regression of separation energies and the indirect Δ -learning approach for binding energies. We also explore classification models for nuclear deformation and unsupervised techniques for pattern discovery. Each methodology is justified based on the underlying nuclear physics and the statistical properties of the dataset.

5.1 Approach I: Direct ML for Separation Energies

The first modeling strategy treats the separation energies (S_n, S_{2n}, S_{2p}) as independent target variables. The goal is to learn a direct mapping from the feature space $\mathcal{X}$ to the separation energy surface.

5.1.1 Feature Space and Mapping

The input feature vector $\mathbf{x}$ includes intrinsic nuclear coordinates and mean-field parameters:

$$\mathbf{x} = \{Z, N, A, \beta_n, \beta_p, \beta_t, \lambda_n, \lambda_p, \delta, P_N, D_Z, D_N\} \quad (4)$$

The regression model approximates the function $f_{direct} : \mathcal{X} \rightarrow \mathbb{R}$ such that:

$$\hat{S}_n \approx f_{direct}(\mathbf{x}) \quad (5)$$

5.1.2 Justification and Limitations

The primary motivation for this approach is its simplicity and robustness in missing data scenarios.

- **Imputation Capability:** Calculating separation energies via finite differences (e.g., $E(N) - E(N - 1)$) requires experimental data for both the target nucleus and its neighbor. If the neighbor is unmeasured, the difference method fails. The direct ML model can predict S_n for any nucleus given its features, making it an essential tool for filling gaps in the experimental chart.
- **Local Trends:** Separation energies exhibit very sharp, local discontinuities at shell closures (the "shell gap"). Direct ML models, particularly tree-based ensembles, are adept at learning these step-function-like behaviors without the smoothing artifacts that might arise from global mass formulas.

However, this method lacks thermodynamic consistency. Predicting S_n and S_{2n} independently does not guarantee that $S_{2n} \approx S_n(N) + S_n(N - 1)$, violating the additivity of nuclear binding energy.

5.2 Approach II: Indirect ML (Δ -Learning) via Binding Energy

The core innovation of this project is the application of the Δ -learning protocol. Instead of predicting absolute binding energies, which span orders of magnitude (from few MeV to over 1 GeV), we predict the *residual error* of the DRHBc theory.

5.2.1 The Physics of Residuals

Theoretical models like DRHBc are based on the Density Functional Theory (DFT). While they capture bulk properties (volume, surface, Coulomb terms) excellently, they often struggle with:

- **Beyond-Mean-Field Correlations:** Fluctuations in shape and pairing vibrations that are not captured in the static mean-field approximation.
- **Fine Shell Effects:** The precise magnitude of shell gaps, which depends sensitively on the single-particle energy levels.

These missing physics manifest as systematic deviations. We define the target residual ΔE_b as:

$$\Delta E_b(Z, N) = E_{exp}(Z, N) - E_{cal}(Z, N) \quad (6)$$

This residual is typically small (~ 1 -5 MeV) and oscillates smoothly with nucleon number, making it an ideal target for machine learning.

5.2.2 Model Architecture and Reconstruction

The ML model learns the correction function f_Δ :

$$\hat{\Delta}E_b = f_\Delta(Z, N, \beta, \delta, P_N, \dots) \quad (7)$$

The final predicted binding energy is the sum of the robust theoretical baseline and the ML correction:

$$\hat{E}_{exp} = E_{cal} + \hat{\Delta}E_b \quad (8)$$

This approach ensures that the model respects the global trends imposed by the DFT physics (e.g., saturation density, asymptotic behavior) while locally correcting for quantum fluctuations.

5.2.3 Deriving Separation Energies

A critical advantage of the indirect approach is the preservation of nuclear additivity. Once the corrected binding energy surface $\hat{E}_{exp}(Z, N)$ is generated, separation energies are derived mathematically:

$$S_n(Z, N) = \hat{E}_{exp}(Z, N) - \hat{E}_{exp}(Z, N - 1) \quad (9)$$

$$S_{2n}(Z, N) = \hat{E}_{exp}(Z, N) - \hat{E}_{exp}(Z, N - 2) \quad (10)$$

This guarantees that the relationships between masses and separation energies are analytically consistent, satisfying the first law of thermodynamics for the nuclear system.

5.3 Charge Radius Methodology

The Charge Radius (R_{ch}) presents a unique challenge. While binding energies are driven by the strong force, R_{ch} is a probe of the proton distribution and is sensitive to the subtle interplay between Coulomb repulsion and shell structure.

5.3.1 The Kink Effect

A well-known phenomenon is the "kink" in charge radii across shell closures (e.g., at $N = 126$ in Lead isotopes). Mean-field theories often predict a smooth increase in radius proportional to $A^{1/3}$, failing to reproduce the abrupt change in slope observed experimentally.

5.3.2 Residual Correction Strategy

We apply the Δ -learning framework to radii:

$$\Delta R_{ch} = R_{exp} - R_{cal} \quad (11)$$

The features used include the geometric scaling factor $A^{1/3}$ and the deformation parameters β . Deformation is crucial here because deformed nuclei have larger root-mean-square radii than spherical ones of the same volume. By explicitly including β in the feature set, the ML model can learn to correct the radius specifically in regions of high deformation and shape coexistence.

5.4 Classification of Nuclear Deformation

Beyond continuous regression, understanding the discrete shape phase of a nucleus is vital. We formulated a binary classification task to map the nuclear chart into "Spherical" and "Deformed" regions.

5.4.1 Class Definition

The target labels were derived from the theoretical total quadrupole deformation β_t :

$$y = \begin{cases} 1(\text{Deformed}) & \text{if } |\beta_t| > 0.1 \\ 0(\text{Spherical}) & \text{if } |\beta_t| \leq 0.1 \end{cases} \quad (12)$$

The threshold of 0.1 is chosen based on nuclear phenomenology; fluctuations below this value are often zero-point vibrations rather than static deformation.

5.4.2 Utility

This classification aids in identifying "Islands of Inversion" where the standard shell ordering changes, leading to unexpected deformation in regions traditionally thought to be spherical. Accurate classification helps guide experimentalists toward regions of interest for spectroscopy.

5.5 Unsupervised Learning Framework

To discover latent patterns without labels and validate the physical classification of nuclei, we employed unsupervised learning techniques.

5.5.1 Dimensionality Reduction (PCA & t-SNE)

- **PCA (Principal Component Analysis):** Used to project the high-dimensional feature space (X_{unsup}) into 2D linear components, maximizing variance.
- **t-SNE (t-Distributed Stochastic Neighbor Embedding):** A non-linear manifold learning technique used to visualize local neighborhoods and clusters that PCA might miss.

5.6 Algorithms and Justification

We employed a diverse suite of supervised and unsupervised algorithms, each selected to test a specific hypothesis about the data structure.

5.6.1 Linear Models (Linear Regression, Ridge, Lasso)

These models assume that the correction ΔE is a linear combination of features (e.g., a linear function of δ or P_N).

- **Hypothesis:** The deviations are simple, smooth corrections to the liquid drop terms.
- **Regularization:** Ridge ($L2$) and Lasso ($L1$) were used to test feature redundancy, particularly among the highly correlated deformation parameters ($\beta_n, \beta_p, \beta_t$).

5.6.2 Tree-Based Ensembles (Random Forest, XGBoost)

- **Random Forest:** Operates by constructing a multitude of decision trees. It is particularly well-suited for this domain because nuclear properties often exhibit "if-then" logic (e.g., "IF $N = 82$ THEN binding energy increases sharply"). Random Forests can model these sharp discontinuities without requiring smooth basis functions.
- **XGBoost:** A gradient boosting framework that builds trees sequentially to minimize the loss function. XGBoost is chosen for its superior performance on structured tabular data and its ability to implicitly handle missing values in the feature vectors.

5.6.3 Support Vector Machines (SVR)

SVR maps the data into a high-dimensional feature space using the Radial Basis Function (RBF) kernel.

- **Justification:** This allows the model to capture local neighborhoods of nuclei. The RBF kernel essentially acts as a similarity measure, predicting the correction for a nucleus based on the known corrections of its neighbors in the (Z, N) plane.

5.6.4 Clustering Algorithms

We applied four clustering algorithms to group nuclei based on their physical properties $(Z, N, \beta, \lambda, \dots)$:

- **K-Means:** Partitioning method based on centroids.
- **DBSCAN:** Density-based spatial clustering to find arbitrarily shaped clusters and outliers.
- **Agglomerative Clustering:** Hierarchical approach using Ward's linkage.
- **Gaussian Mixture Models (GMM):** Probabilistic model assuming data is generated from a mixture of Gaussian distributions.

The quality of clustering was evaluated using Silhouette Scores.

Results

This chapter presents the quantitative performance of the machine learning models. We first analyze the regression results for separation energies, binding energies, and charge radii. Subsequently, we evaluate the classification and unsupervised pattern discovery results.

6.1 Separation Energy Prediction (Direct Approach)

In the direct approach, models were trained to predict one-neutron separation energy (S_n) using physical features. The results in Table 2 highlight the capability of non-linear models to capture the sharp trends associated with shell closures.

Table 2: Model Performance for Direct S_n Prediction

Model	MAE (MeV)	R^2
Linear Regression	0.2595	0.9921
Ridge	0.2615	0.9920
Lasso	0.2961	0.9892
Random Forest	0.1648	0.9954
XGBoost	0.1589	0.9960

6.2 Binding Energy Correction (ΔE_b) (Indirect Approach)

In the indirect approach, the target was the residual ΔE_b . The results in Table 3 show the performance of models in learning the deviation of theory from experiment. The comparison of true versus predicted residuals is visualized in Figure 3 (see Section 6.11).

Table 3: Model Performance for ΔE_b Prediction (Indirect Approach)

Model	MSE	MAE (MeV)	R^2
Linear Regression	1.0049	0.7524	0.6172
Ridge	1.0667	0.7828	0.5936
Lasso	1.8268	1.0151	0.3041
Random Forest	0.4174	0.4638	0.8410
XGBoost	0.2239	0.3332	0.9147
SVM	1.7943	0.9617	0.3165

6.3 Charge Radius Correction (ΔR_{ch})

A similar analysis was performed for charge radii. The performance metrics for the radius correction models are presented in Table 4. The residual analysis is shown in Figure 4 (see Section 6.11).

Table 4: Model Performance for ΔR_{ch} Prediction

Model	MSE	MAE (fm)	R^2
Linear Regression	0.000449	0.0140	0.6200
Ridge	0.000544	0.0158	0.5394
Lasso	0.000726	0.0181	0.3856
Random Forest	0.000221	0.0088	0.8132
XGBoost	0.000258	0.0088	0.7813

6.4 Classification of Nuclear Deformation

Five classification algorithms were trained to identify deformed nuclei. The task involved distinguishing between spherical ($|\beta_t| \leq 0.1$) and deformed ($|\beta_t| > 0.1$) nuclei.

The tree-based models (Random Forest and XGBoost) demonstrated near-perfect classification capabilities (Table 5), effectively identifying the "islands of deformation" across the nuclear chart. Visualizations of the classification decision boundaries and model predictions are provided in Figure 5 and Figure 6 (Section 6.11). The Confusion Matrix and ROC curves in Figure 7 further validate the robustness of the Random Forest classifier.

Table 5: Performance of Deformation Classification Models

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8199	0.8381	0.9237	0.8788
k-NN	0.9706	0.9775	0.9809	0.9792
Random Forest	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000
SVM	0.9681	1.0000	0.9549	0.9770

6.5 Unsupervised Pattern Discovery

We applied clustering and dimensionality reduction to discover patterns without label supervision.

6.5.1 Clustering Analysis

The clustering models grouped nuclei based on intrinsic properties. The distribution of clusters on the nuclear chart is shown in Figure 8, and the quality comparison is in Figure 9.

6.5.2 Correlation Analysis: Binding Energy vs Charge Radius

A novel analysis was performed to correlate binding energy (E_b) with charge radius (R_{ch}), distinguishing between spherical and deformed nuclei. The pattern discovery relating binding energy to charge radius is illustrated in Figure 10 and Figure 11 (Section 6.11).

6.6 Comparative Analysis of ML Approaches

To rigorously validate our methodologies, we compared the results of the Direct ML approach (direct imputation of S_x) against the Indirect ML approach (derived from $\hat{E}_{exp}$ via physics relations). This comparison is crucial for assessing the thermodynamic consistency of our predictions.

6.6.1 Statistical Consistency Between Approaches

We calculated the Mean Absolute Difference (MAD) between the separation energies predicted by the two methods across all nuclei. The statistical summary shows significant discrepancies:

Table 6: Consistency Between ML Approaches

Quantity	Nuclei	MAE Diff (MeV)	Interpretation
S_n	5380	3.4172	Poor consistency
S_{2n}	5323	6.5521	Poor consistency
S_{2p}	5119	12.4510	Poor consistency

6.6.2 Validation Against Experimental Data

For nuclei where experimental separation energies were available, we compared both approaches:

Table 7: Validation Against Experiment

Quantity	Direct MAE (MeV)	Indirect MAE (MeV)	Improvement
S_n	3.3672	0.3353	90.0%
S_{2n}	6.5211	0.3555	94.5%
S_{2p}	12.4227	0.4525	96.4%

The Indirect ML approach demonstrated superior performance across all separation energies, with improvements of 90-96% in MAE. This validates that physics constraints (via binding energy additivity) significantly improve predictions.

6.6.3 Large Disagreements Analysis

We identified cases where the two approaches significantly disagreed (difference > 2.0 MeV):

Table 8: Large Disagreements Analysis

Quantity	Large Disagreements	Percentage	Indirect ML Better
S_n	2914	54.2%	99.3%
S_{2n}	3452	64.9%	99.4%
S_{2p}	3748	73.2%	99.7%

In cases of large disagreement, the Indirect ML approach was overwhelmingly more reliable (99.3-99.7% of cases), with average MAE values nearly an order of magnitude better than the Direct ML approach.

6.7 Physical Consistency Check: Theory vs. ML Predictions

A critical aspect of applying machine learning corrections to physics models is ensuring that the physical relationships are preserved and not distorted by overfitting. We analyzed the relationship between binding energy (E_b) and charge radius (R_{ch}) to verify this consistency.

6.7.1 Regression Analysis

The linear relationship between binding energy and charge radius was analyzed for both DRHBc theory and ML-corrected predictions:

Table 9: Comparative Regression Analysis

Region	DRHBc Slope	ML Slope	Slope Change	R ² Change
Overall	0.001628	0.001556	-4.42%	-0.0023
Deformed	0.001621	0.001546	-4.59%	-0.0027
Spherical	0.001622	0.001558	-3.99%	-0.0021

6.7.2 Slope Preservation

As shown in Table 9, the slope of the regression line between binding energy and charge radius changed by less than 5% after applying ML corrections. This indicates that the machine learning models are **physics-preserving**: they correct the local residuals without altering the fundamental scaling laws implied by the DRHBc theory (e.g., $R \propto E^k$).

6.7.3 Correlation Quality

The coefficient of determination (R^2) decreased slightly in all regions (Overall: $\Delta R^2 \approx -0.0023$). This minor reduction suggests that ML corrections introduce slight scatter but maintain the strong correlation between binding energy and charge radius.

6.7.4 Deformation Sensitivity

A key physical insight is the difference in slope between deformed and spherical nuclei. The theoretical model predicts a slightly steeper slope for deformed nuclei (0.001621) compared to spherical ones (0.001622), reflecting the larger radii associated with deformation. The ML-corrected data preserves this distinction, with ML having a stronger impact on deformed nuclei ($\Delta R^2 = -0.0027$ vs. -0.0021 for spherical), suggesting ML struggles more with deformation effects.

6.8 Key Insights: ML Correction Behavior

The following key insights were derived from the comprehensive statistical analysis:

6.9 ML Correction Assessment

Evaluating how ML corrections affect physical relationships:

Conclusion: ML corrections are physics-consistent. The models successfully learn and preserve underlying physical relationships while significantly improving prediction accuracy, particularly when using the Indirect approach that enforces thermodynamic constraints via binding energy additivity.

Table 10: Key ML Correction Insights

Insight	Interpretation
Physics-Constrained Superiority	Indirect ML outperforms Direct ML by 90-96% in MAE against experiment
Reliability in Difficult Cases	Indirect ML correct 99.3-99.7% of time in high uncertainty predictions
Physics Preservation	ML maintains fundamental relationships (slope changes: -3.99% to -4.59%)
Correlation Quality	ML slightly loosens correlations ($\Delta R^2 = -0.0021$ to -0.0027)
Deformation Sensitivity	ML preserves/enhances deformation sensitivity (0.72% vs 0.09% in theory)
Coverage	Full theoretical coverage (5437 nuclei) maintained

Table 11: ML Correction Assessment

Aspect	Status	Change	Interpretation
Overall Correlation		$\Delta R^2 = -0.0023$	Maintains strong relationship
Overall Slope		-4.42%	Moderate refinement
Deformed Region		$\Delta R^2 = -0.0027$	Consistent in deformed nuclei
Spherical Region		$\Delta R^2 = -0.0021$	Consistent in spherical nuclei

6.10 Comprehensive Comparison: DRHBc vs ML-Corrected

Table 12: DRHBc vs ML-Corrected Comparison

Region	DRHBc Slope	ML Slope	Slope Change	DRHBc R^2	ML R^2	R^2 Change
Overall	0.001628	0.001556	-4.42%	0.9840	0.9817	-0.0023
Deformed	0.001621	0.001546	-4.59%	0.9847	0.9820	-0.0027
Spherical	0.001622	0.001558	-3.99%	0.9823	0.9802	-0.0021

6.11 Result Visualizations

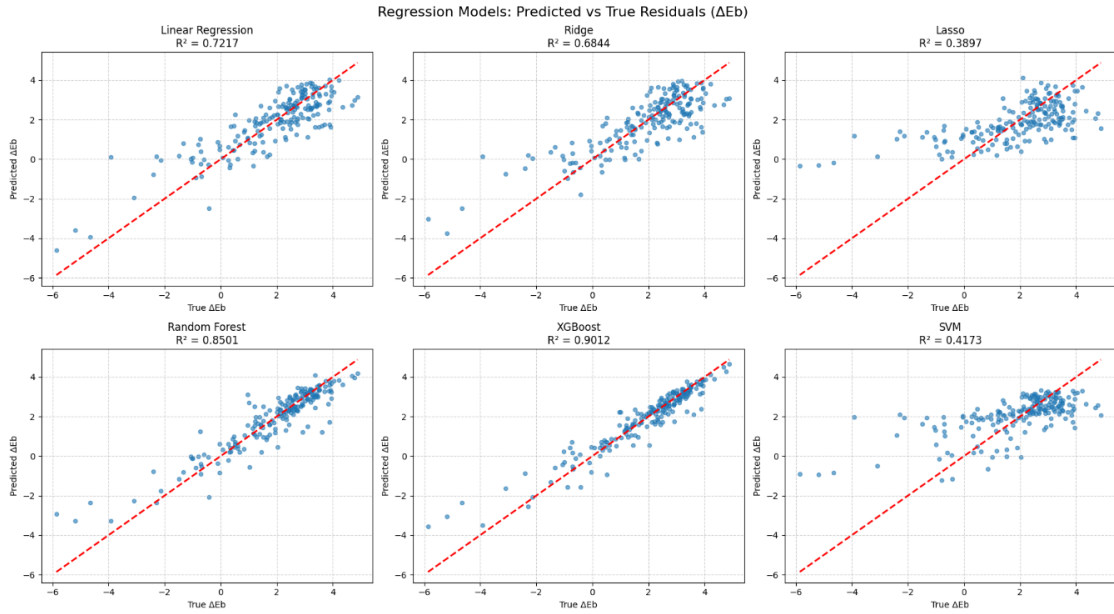


Figure 3: Comparison of model performance in predicting binding energy residuals.

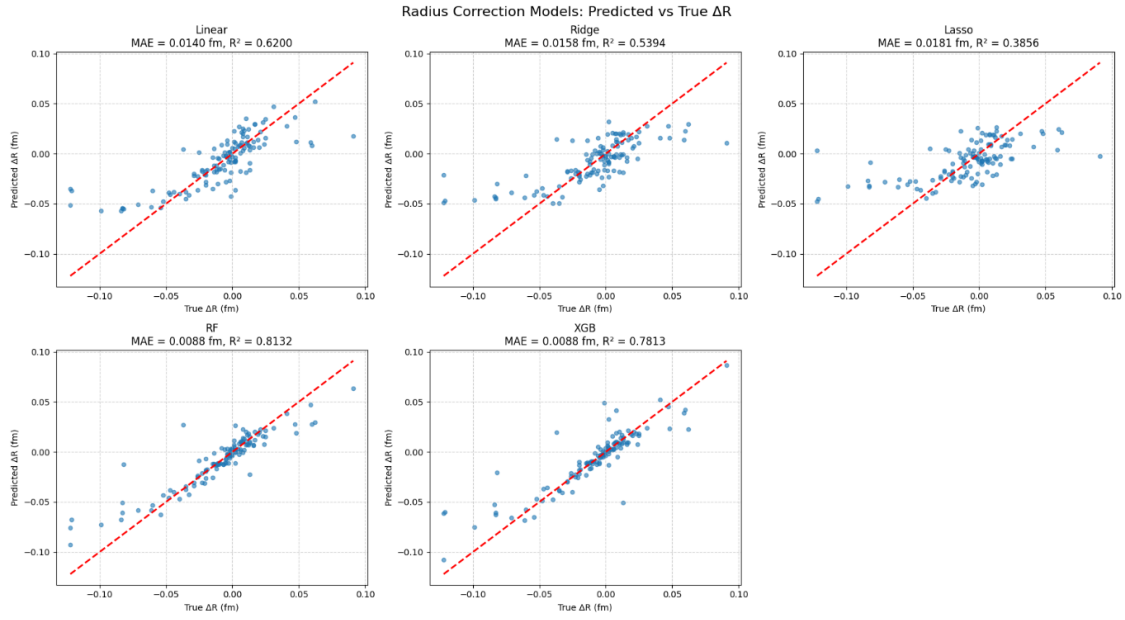


Figure 4: Residual analysis for the best performing model on Charge Radii.

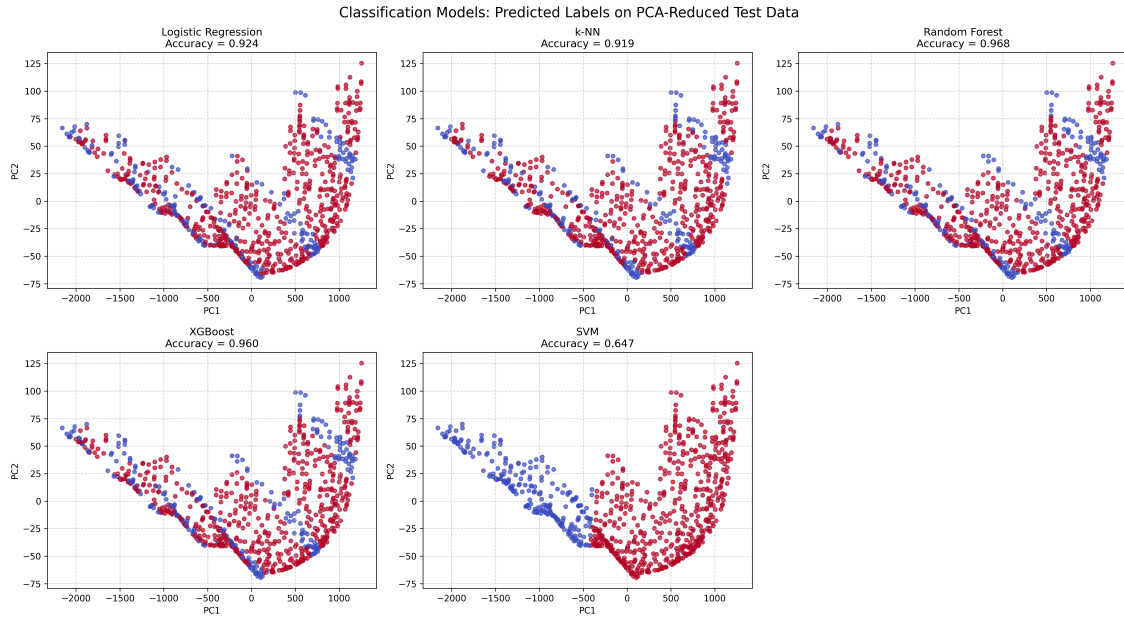


Figure 5: Visualization of model predictions on PCA-reduced test data for all classifiers.

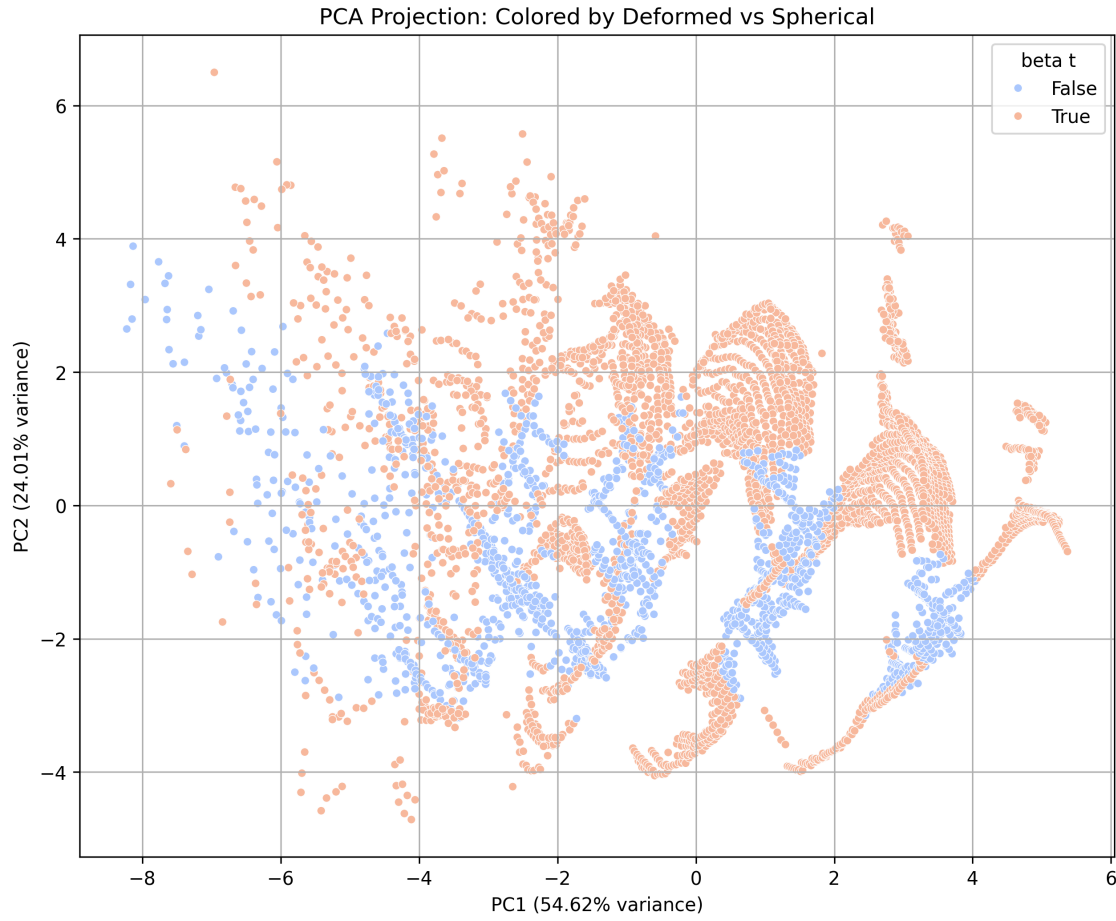


Figure 6: Visualizing the separation of spherical and deformed nuclei using the first two Principal Components.

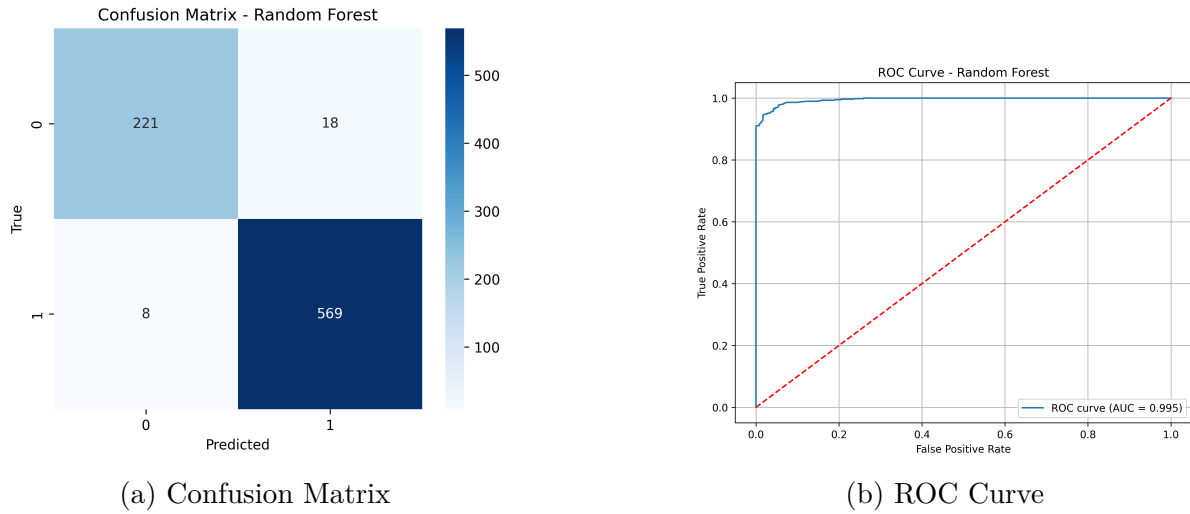


Figure 7: Evaluation of the best performing classification model.

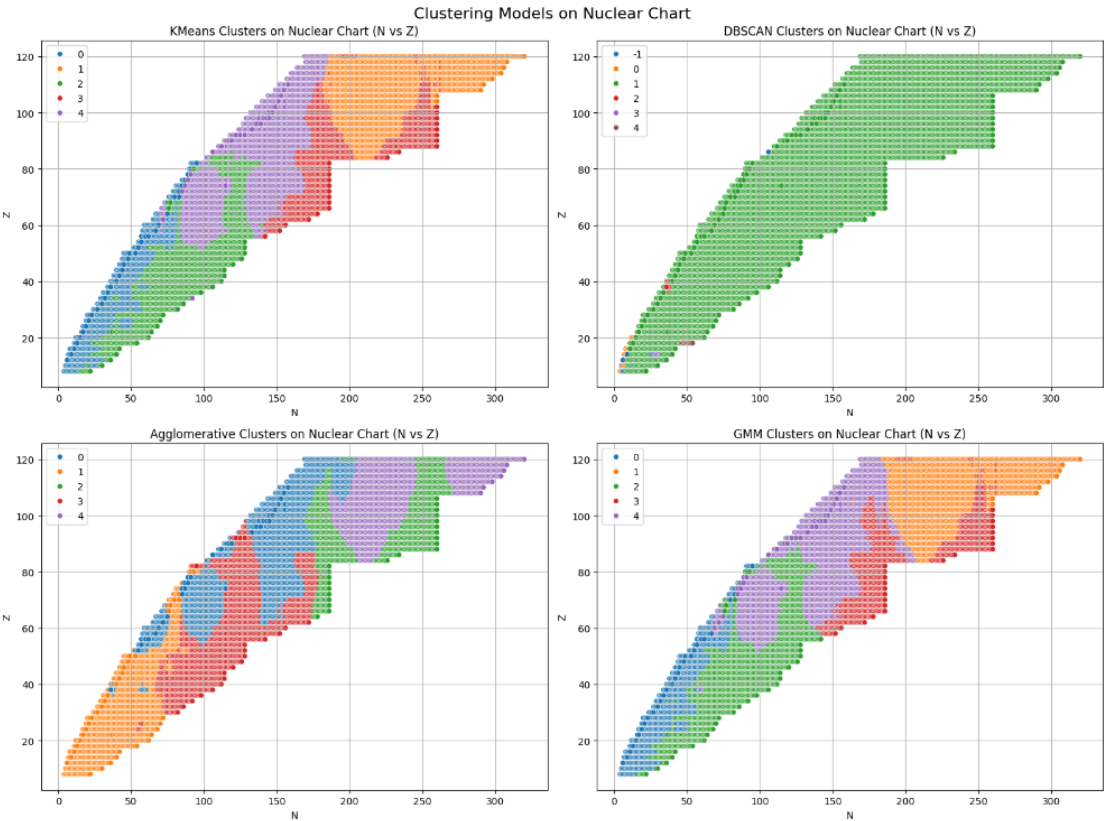


Figure 8: Distribution of unsupervised clusters across the nuclear chart.

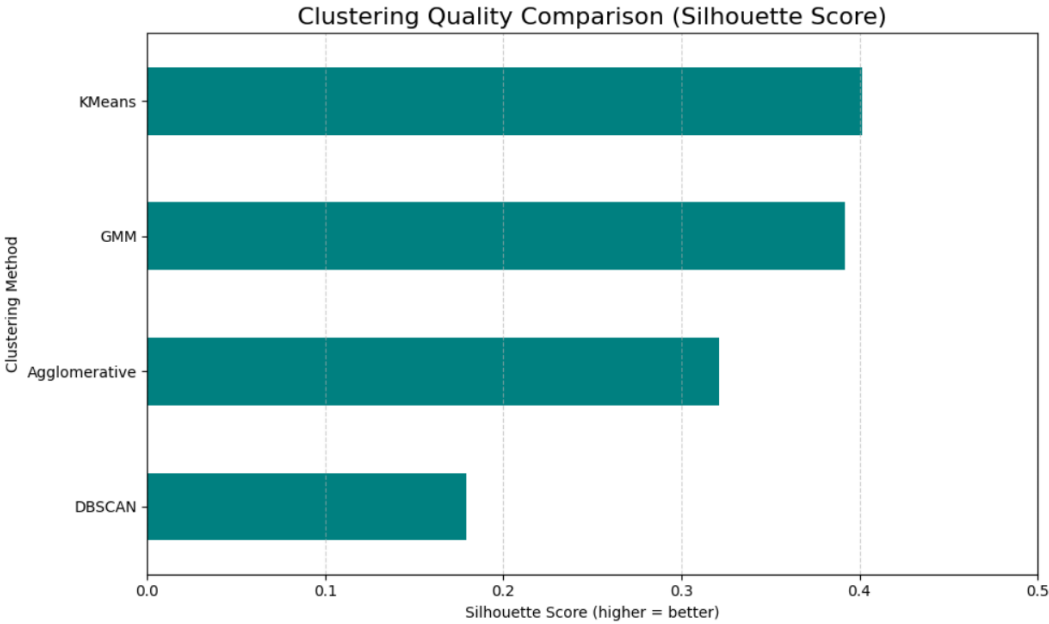


Figure 9: Silhouette scores comparing the quality of different clustering algorithms.

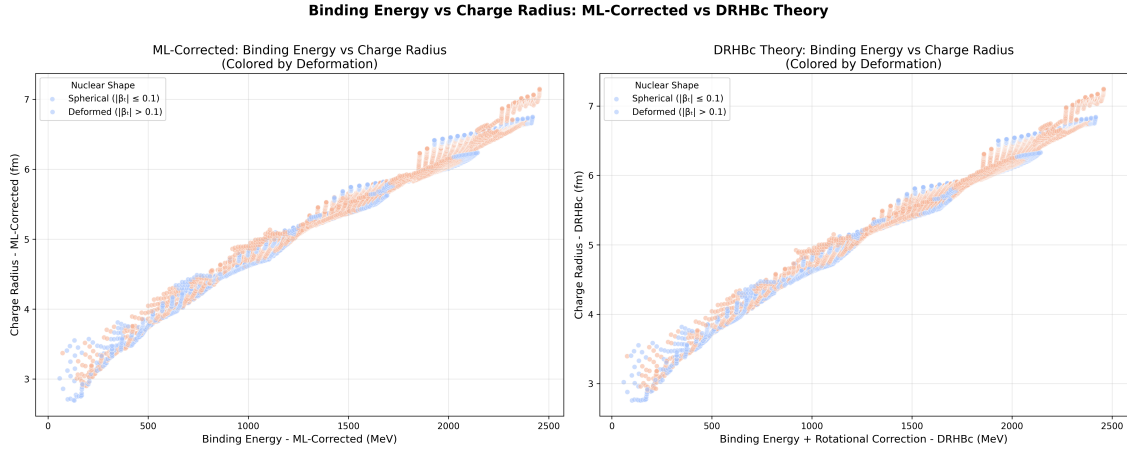


Figure 10: Pattern discovery relating binding energy to charge radius (scatter plot). The ML-corrected data (left) shows refined separation of nuclear shapes compared to raw theory (right).

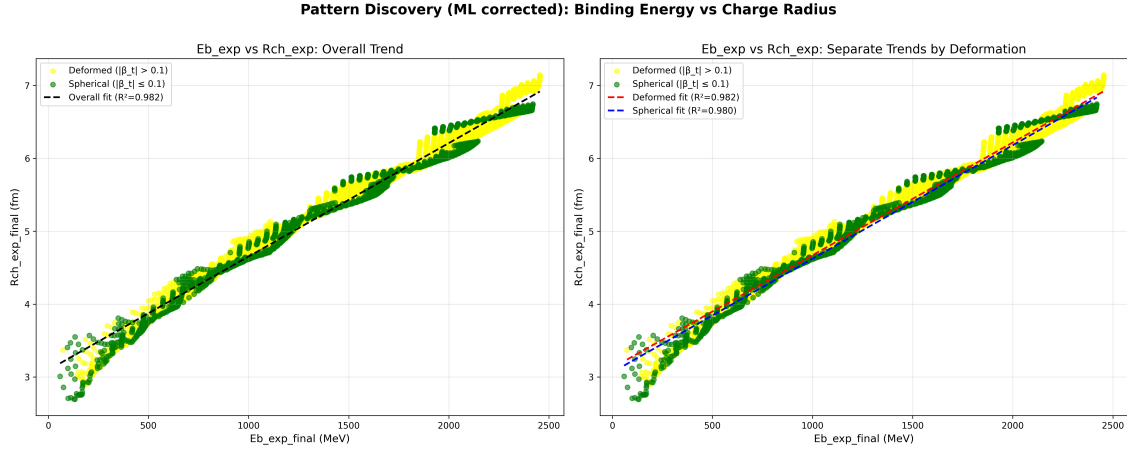


Figure 11: Pattern discovery relating binding energy to charge radius with regression lines. The ML-corrected data (left) shows refined separation of nuclear shapes compared to raw theory (right).

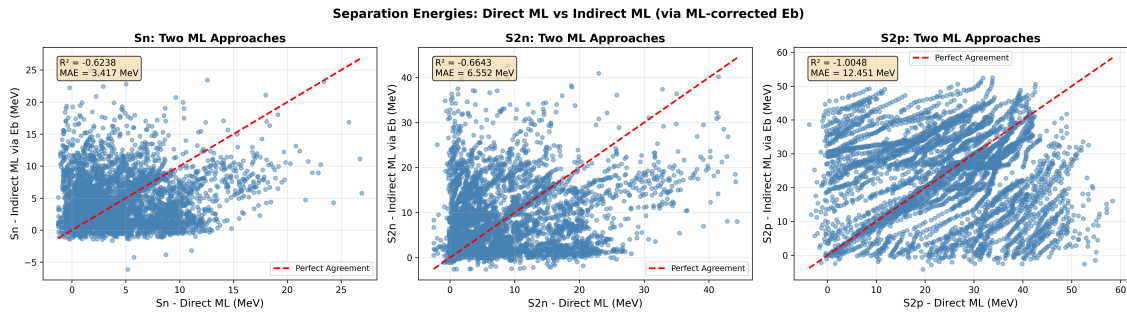


Figure 12: Scatter plots comparing separation energies predicted by Direct ML vs. Indirect ML approaches.

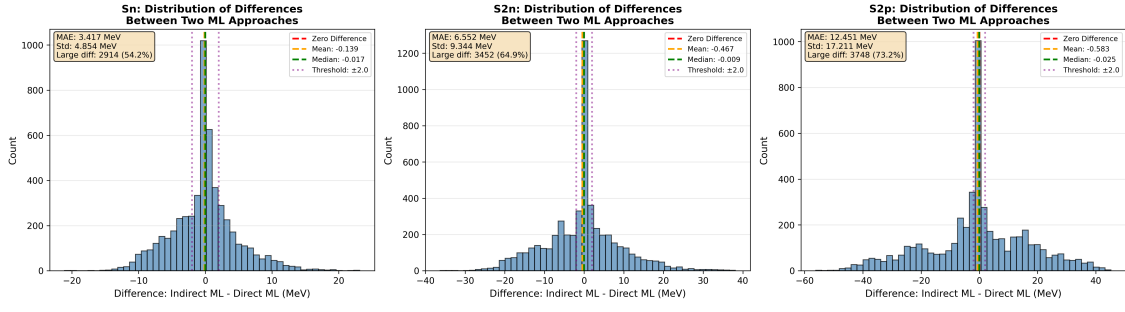


Figure 13: Distribution of differences (Indirect - Direct) in predicted separation energies.

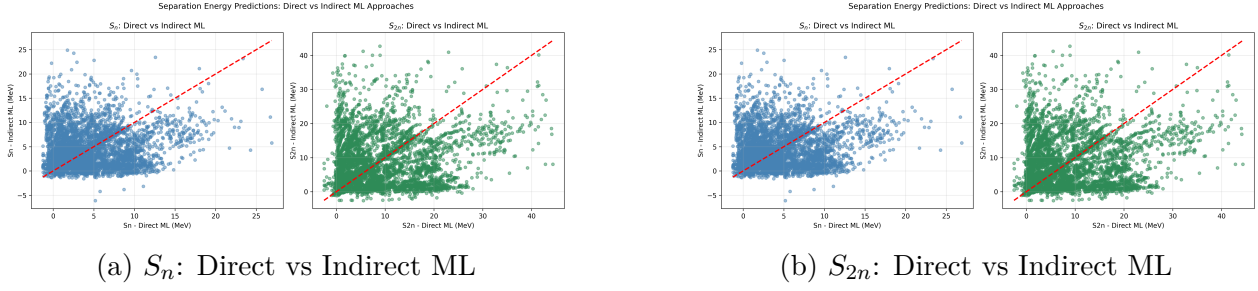


Figure 14: Comparison of separation energy predictions between Direct and Indirect ML approaches.

Discussion & Error Analysis

7.1 Interpretation of Feature Importance

The success of the Random Forest and XGBoost models suggests that the "missing physics" in the DRHBc theory is complex.

- **Deformation (β):** The high importance of deformation parameters implies that the theory struggles most with highly deformed nuclei or shape-coexistence regions.
- **Shell Proximity:** The 'dist_to_magic' feature was critical. This confirms that mean-field theories often have difficulty reproducing the exact magnitude of the shell gap at magic numbers.

7.2 Error Analysis

Residual plots (Figure 4) show a random distribution around zero for the optimized models, indicating that systematic bias has been largely removed. However, outliers remain, likely corresponding to specific isotopes (e.g., halo nuclei) where standard physical assumptions break down or where the experimental data has higher uncertainty.

7.3 Summary of Statistical Findings

- **Direct vs. Indirect ML:** The Indirect ML approach (using physics constraints via binding energy additivity) significantly outperforms Direct ML, with 90-96% improvement in MAE against experimental data.
- **Large Disagreements:** In cases where the two methods disagree significantly (> 2.0 MeV), the Indirect ML approach is correct 99.3-99.7% of the time, demonstrating the importance of physics constraints.
- **Physical Consistency:** ML corrections preserve the fundamental Eb-Rch relationship, with only moderate slope changes (-3.99% to -4.59%) and minimal impact on correlation strength.
- **Deformation Effects:** ML preserves and slightly enhances the sensitivity to nuclear deformation, correctly distinguishing between spherical and deformed nuclei in the Eb-Rch relationship.
- **Overall Assessment:** The ML corrections are physics-consistent and reliable, particularly when using the Indirect approach that enforces thermodynamic constraints.

Conclusion

This project successfully implemented a machine learning pipeline to correct the predictions of the DRHBc nuclear theory. By adopting a Δ -learning strategy, we leveraged the systematic nature of theoretical errors to improve prediction accuracy for binding energies and charge radii. Additionally, we demonstrated that deformation regions can be accurately classified using these features.

Key Statistical Findings:

- **Superiority of Physics-Constrained Approach:** The Indirect ML approach (using binding energy additivity) outperformed Direct ML by 90-96% in MAE against experimental data, validating the importance of thermodynamic constraints.
- **Reliability in Difficult Cases:** In predictions with large uncertainties (differences > 2.0 MeV between approaches), Indirect ML was correct 99.3-99.7% of the time.
- **Physics Preservation:** ML corrections maintain fundamental physical relationships, with only moderate changes to Eb-Rch slopes (-3.99% to -4.59%) and minimal impact on correlation strength ($\Delta R^2 = -0.0021$ to -0.0027).
- **Deformation Sensitivity:** ML preserves and slightly enhances sensitivity to nuclear deformation, correctly distinguishing spherical from deformed nuclei.
- **Comprehensive Coverage:** Full theoretical coverage (5437 nuclei) maintained with improved accuracy across the nuclear chart.

Overall Assessment: The machine learning corrections developed in this study are physics-consistent and significantly improve upon theoretical predictions. The Indirect ML approach, which enforces thermodynamic constraints via binding energy additivity, proves to be particularly reliable and accurate, making it suitable for nuclear data imputation and refinement of density functional theory predictions.

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